

1. Consider the definite integral $\int_1^3 3x \, dx$.
 - a) Sketch what is graphically represented by this integral.
 - b) Use geometry to calculate this integral.
 - c) Show that this integral equals 12 using the limit definition of the definite integral.
 - d) Show that this integral equals 12 using the limit definition of the definite integral and where you use left endpoints for your selection of x_i^* .
 - e) TRUE or FALSE: If the function $y = 3x$ is integrable on $[1, 3]$ then it must always produce an integral value of 12 regardless of the selection of the points for x_i^* .
 - f) Based on your answer to the previous part, how could ever possibly show that a function is *NOT* integrable?

2. Evaluate the integral $\int_0^1 (x^3 - 3x^2) \, dx$ using the limit definition of the definite integral.

3. Relate each of the following limits to a definite integral over the specified interval. (Use right endpoints.)

- a) $\lim_{n \rightarrow \infty} \frac{\sqrt{1} + \sqrt{2} + \sqrt{3} + \dots + \sqrt{n}}{n\sqrt{n}}$ over $[0, 1]$.

- b) $\lim_{n \rightarrow \infty} \pi \left(\frac{\sin\left(\frac{2\pi}{n}\right) + \sin\left(\frac{4\pi}{n}\right) + \sin\left(\frac{6\pi}{n}\right) + \dots + \sin\left(\frac{2n\pi}{n}\right)}{n} \right)$ over $[0, \pi]$.

- c) $\lim_{n \rightarrow \infty} \frac{1^7 + 2^7 + 3^7 + \dots + n^7}{n^8}$ over $[0, 1]$.

4. Consider the integral $\int_1^2 x^{-2} \, dx$.

- a) Construct a limit (where x_i^* represents the right endpoints of our sub-intervals) that is equal to this integral.
- b) In the previous questions we were always using x_i^* as the right endpoint of our intervals. This worked well for these examples, but sometimes it can be easier to use other values. Show how if we choose x_i^* to be the geometric mean of x_{i-1} and x_i (that is, $x_i^* = \sqrt{x_{i-1} \cdot x_i}$) then we can find $\int_1^2 x^{-2} \, dx$.

Hint: determine some formulas for x_{i-1} & x_i and use the identity $\frac{1}{m(m+1)} = \frac{1}{m} - \frac{1}{m+1}$.